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Question Paper

MANIPAL ACADEMY OF HIGHER EDUCATION

III Semster Mid Term Examination September 2025

DIGITAL SYSTEMS AND COMPUTER ORGANIZATION [CSS 2104]

Marks: 30 **Duration: 90 mins.**

MCQ

Answer all the questions. Section Duration: 20 mins

1) What is the minimum number of 8-to-1, 4-to-1, and 2-to-1 multiplexers required to implement a 32-to-1 multiplexer? (0.5)

4 of 8-to-1, 1 of 4-to-1, 0 of 2-to-1 | 2 of 8-to-1, 2 of 4-to-1, 1 of 2-to-1 | 8 of 4-to-1, 2 of 2-to-1 | 1 of 8-to-1, 4 of 4-to-1, 1 of 2-to-1

2) What is the output of a 3-to-8 decoder when the input is 101? (0.5)

Output line 5 is high. | Output line 2 is high. | Output line 6 is high. | Output line 7 is high.

3) In a priority encoder, if inputs D3 = 1, D2 = 1, D1 = 0, D0 = 1, with D3 having the highest priority, what will be the output? (0.5)

00 | 01 | 10 | 11

4) Which of the following is used to detect overflow in binary addition? (0.5)

Carry into MSB | Carry out of MSB | XOR of carry into and carry out of MSB | AND of MSB bit

5) Which of the following is a valid way to declare a 4-bit register in Verilog? (0.5)

reg[4] a; | reg[3:0] a; | register[3:0] a; | wire[4:1] a;

6) A SR latch is _____ (0.5)

Combinational circuit | Synchronous sequential circuit | 1 bit memory element | 1-clock delay element

7) The characteristic equation of MN- flipflop is (0.5)

M	N	Output
0	0	Toggle
0	1	1
1	0	0
1	1	No change

$Q(t+1) = MQ^I + N^IQ$ | $Q(t+1) = M^IQ^I + NQ$ | $Q(t+1) = MQ+N^IQ^I$ | $Q(t+1) = M^IQ+NQ^I$

8) Identify the TRUE statement/s with respect to Verilog code from the following: (0.5)

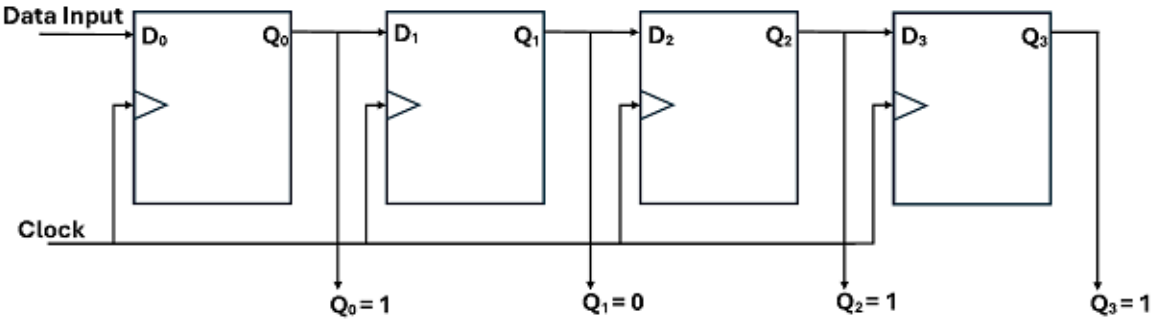
i) The procedural statements should appear inside the always block

ii) A sub-circuit can be instantiated inside the always block

iii) A subcircuit cannot be instantiated if the order of signals listed in the instantiation statement is not same as the order of the ports given in the module statement of the subcircuit.

i) and ii) only | iii) only | i) only | ii) and iii) only

9) (0.5)



The above figure depicts a 4-bit Serial-in-Parallel-out (SIPO) shift register that currently holds the value 1101. Q_3 is the Most Significant Bit (MSB) and Q_0 is the Least significant Bit (LSB). If a new value 0110 is supplied serially through the *Data Input*, what will be the content (as Q_3 Q_2 Q_1 Q_0) of the above register after three consecutive clock cycles? (Note: the serial input starts with the MSB and ends with the LSB of 0110)

1011 | 1101 | 0111 | 0110

10) Assume a 4-bit Johnson counter holds an initial value of 1000. Which of the following depicts the behaviour of a Johnson counter after applying three consecutive clock signals (Clk)? (0.5)

Initial 1 0 0 0 Clk1 0 0 0 1 Clk2 1 0 0 0 Clk3 0 0 0 1 | Initial 1 0 0 0 Clk1 0 1 0 0 Clk2 0 0 1 0 Clk3 0 0 0 1 | Initial 1 0 0 0 Clk1 1 1 0 0 Clk2 1 1 1 0 Clk3 1 1 1 1 | Initial 1 0 0 0 Clk1 0 1 0 0 Clk2 1 0 0 0 Clk3 0 1 0 0

DESCRIPTIVE

Answer all the questions.

11) Derive the expressions for the flip flop inputs of a sequential circuit with two *JK* flip-flops *A* and *B* and two inputs *E* and *F* . If $E = 0$, the circuit remains in the same state regardless of the value of *F* . When $E = 1$ and $F = 1$, the circuit goes through the state transitions from 00 to 01, to 10, to 11, back to 00, and repeats. When $E = 1$ and $F = 0$, the circuit goes through the state transitions from 00 to 11, to 10, to 01, back to 00, and repeats. (4)

12) Multiply $(1011)_2$ and $(1101)_2$ using an array multiplier approach. Show the step by step operation to generate the final product. (3)

13) Write Verilog code for 2 to 1 multiplexer using conditional operator. Design (3)

f(a, b, c)=∑(2,3,5,6) using two 2 to 1 multiplexers and one inverter.

14) Explain the working of edge triggered D Flip Flop with three crosscoupled NAND latches with a neat diagram and all the four cases of the inputs D and Clock. (3)

15) Explain why the SR NAND latch is called an active-low latch, and also draw the circuit diagram and truth table of an SR latch using NOR gates. (3)

16) Design a minimal NAND-NAND circuit realization for the expression, assuming that the variables are available in both complemented and uncomplemented form

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$$F(A,B,C,D) = \sum m(1,2,7,8,14) + d(0,6,9,10,15)$$ (3)

17) Design a circuit to obtain 2's complement of a 4 bit number using full adders and inverters. (2)

18) Wrtite the truth table for 8 -to-3 priority encoder. Assume that the input is W[7:0] with W[0] having highest priority . (2)

19) Illustrate the method of combining two 2-to-4 decoders to design a 3-to-8 decoder. (2)

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